

Write-up (Summary)

- **Goal:** Analyze Q4 2020 Australia apparel sales to guide Sales & Marketing on where to invest, which demographics to target, and when to run offers.
- **Data:** `AusApparelSales4thQrt2020.csv` (7,560 rows; fields: Date, Time, State, Group, Unit, Sales). No missing/negative values.
- **Method:** Cleaned whitespace; parsed dates; engineered Month/Week/Quarter; min-max normalized Unit & Sales; GroupBy aggregations; visualized with Seaborn/Matplotlib (state revenue & units, group heatmap, time-of-day bars/heatmap, daily/weekly/monthly/quarterly trends, box plot, distribution).
- **Key findings:**
 - Top revenue states: VIC, NSW, SA; lowest: WA, NT, TAS.
 - Time-of-day: Morning/Afternoon slightly lead; Evening close behind.
 - Demographics: All groups contribute similarly; VIC dominates every group.
- **Recommendations:**
 - Double down on VIC/NSW high-value segments; premium/loyalty offers.
 - Target WA/NT/TAS with localized promos and assortments tuned to their leading group (see heatmap).
 - Staff/ads heavier in Morning/Afternoon; use off-peak for loyalty nudges.
 - Maintain monthly dashboards; monitor weekly for campaign effects.
- **Tools & libs:** Python 3.9, pandas, seaborn, matplotlib, scikit-learn (MinMaxScaler), nbconvert.